

1(a)	I ref. to resolution any <i>one</i> from: Fig. 1.1A is scanning (EM), qualified ; e.g. shows, surface / contour / 3-D, views shows (only the), the outer / external, view (of microvilli) idea of good focus with different depths does not have stated feature of TEM (see mp2) Fig. 1.1B is transmission (EM), qualified ; e.g. showing, 2D / flat, views (of, microvilli / mitochondria) image in one plane showing section through, mitochondria / microvilli / cell I thin shows, internal view of cell / contents of cell / AW does not have stated feature of SEM (see mp1) scanning (EM) stated plus feature <u>and</u> transmission (EM) stated plus feature } ; if Fig. 1.1, A / B not identified	1
1(b)	any <i>two</i> from: max 1 if stating to measure one of the sides of the figure (for each image) measure a number of microvilli and calculate a mean ; same units for actual and measured images / example given ; e.g. <i>if microvillus measurement is in mm</i> convert, microvillus measured length to μm / actual length to mm R if extra detail given is incorrect e.g. <u>divide</u> by 1000 for mm to μm divide, microvillus / image, length by, actual length / AW / $1(\mu\text{m})$; A idea that value for, microvillus / image, length is the magnification	2
1(c)(i)	any <i>two</i> from: (movement of substances) against concentration gradient / from low(er) to high(er) concentration ; (protein has) binding site(s) ; specific ; A description e.g. shape complementary to shape of substance <i>can be in context of protein or binding site</i> ref. to (binding causes protein), conformational change / change of shape ;	2

Question	Answer	Marks
1(c)(ii)	any two from: 1 glucose is moving against a (concentration) gradient ; AW e.g. low to high 2 movement powered by inward movement of Na⁺ ; AW 3 Na⁺ movement (into cell from gut lumen) down an electrochemical gradient or down a (Na⁺), concentration / diffusion, gradient or by facilitated diffusion ; A diffusion if response includes <i>ref. to</i> the cotransport protein / cotranporter 4 AVP ; e.g. <i>idea that</i>, cotransport needed so that <u>all</u> glucose will be removed from gut lumen ora facilitated diffusion of glucose will only be to equilibrium / AW cotransporter has (specific) binding site for glucose I cotransporter has a complementary shape <i>idea that</i> there is no specific active uptake protein for glucose	2
1(c)(iii)	increases / larger / large, surface area (over which glucose can enter cell per unit time) ; A increases / higher / high, SA:V <i>idea of</i> (so) more, transport / carrier / cotransport, proteins A more cotransporters or more proteins, located in membrane / AW ; R pump proteins A <i>idea of</i> more space / places, where proteins can transport glucose	2

Question	Answer	Marks											
1(d)	any four from: <table><tr><th>suggestion max 3</th><th>explanation</th></tr><tr><td> 1 to replace, old / dead / damaged / injured, cells or to, repair / replace, damaged tissue ; I repair cells I replace old tissues </td><td rowspan="2"> 6 idea of <u>function</u> in context of mp1 or 2 or 3 ; e.g. repair allows tissues to function specialisation / differentiation, allows, cells / tissue, to function genetically identical so cells can function as a tissue look for other approaches e.g. 'replace damaged cells that no longer function' is mp1 and mp6 </td></tr><tr><td> 2 cells are produced that can, differentiate / become specialised (into intestinal cells); I multipotent / AW R if, stated / strongly implied that, <u>all</u> cells produced by mitosis become specialised </td></tr><tr><td> 3 produce cells that are genetically identical ; A clones A identical / same, DNA / genetic material / genetic information </td><td> 7 detail of function (intestines) ; e.g. absorption (of, glucose / nutrients) production of digestive enzymes pass absorbed, nutrients / AW, to blood I nutrition </td></tr><tr><td> 4 growth ; R growth of cells </td><td> 8 idea of development during, infancy / childhood ; </td></tr><tr><td> 5 renew / replenish / restore number of / AW, stem cells ; I increase in stem cells </td><td> 9 for, continued production / constant supply / AW, of intestinal cells or so that number of stem cells doesn't decrease ; </td></tr></table>	suggestion max 3	explanation	1 to replace, old / dead / damaged / injured, cells or to, repair / replace, damaged tissue ; I repair cells I replace old tissues	6 idea of <u>function</u> in context of mp1 or 2 or 3 ; e.g. repair allows tissues to function specialisation / differentiation, allows, cells / tissue, to function genetically identical so cells can function as a tissue look for other approaches e.g. 'replace damaged cells that no longer function' is mp1 and mp6	2 cells are produced that can, differentiate / become specialised (into intestinal cells); I multipotent / AW R if, stated / strongly implied that, <u>all</u> cells produced by mitosis become specialised	3 produce cells that are genetically identical ; A clones A identical / same, DNA / genetic material / genetic information	7 detail of function (intestines) ; e.g. absorption (of, glucose / nutrients) production of digestive enzymes pass absorbed, nutrients / AW, to blood I nutrition	4 growth ; R growth of cells	8 idea of development during, infancy / childhood ;	5 renew / replenish / restore number of / AW, stem cells ; I increase in stem cells	9 for, continued production / constant supply / AW, of intestinal cells or so that number of stem cells doesn't decrease ;	4
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